

Course: Networked Control

Part II: Decentralized and distributed control - Introduction

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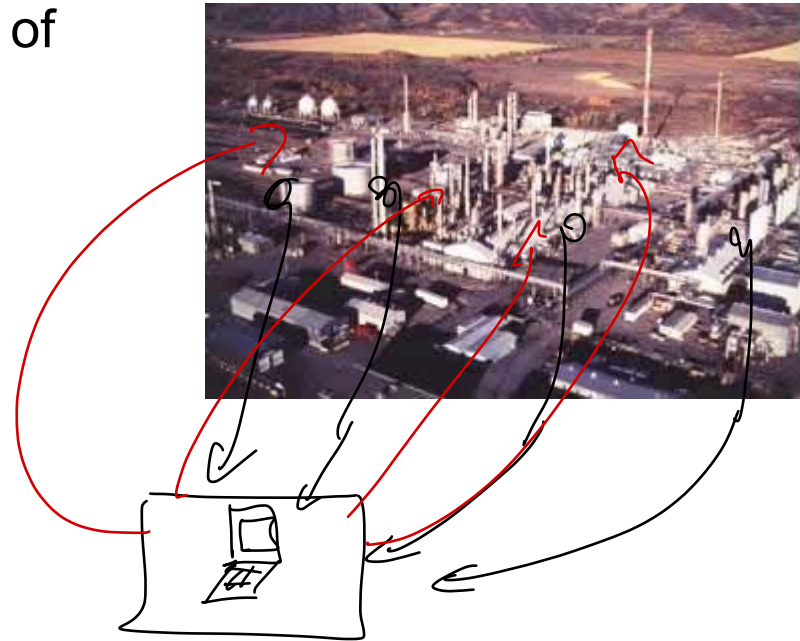
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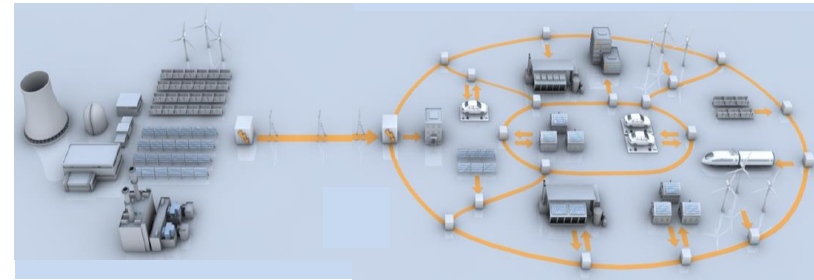
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- **Electric grids**



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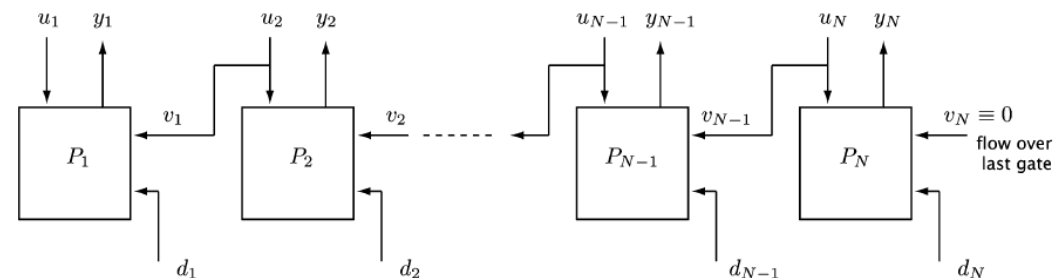
- **Natural gas treatment plants:** interaction of separators, heat exchangers, chillers, mixers, separation columns, ...
- **Electric grids**
- **Fleets of underwater autonomous vehicles**



Examples

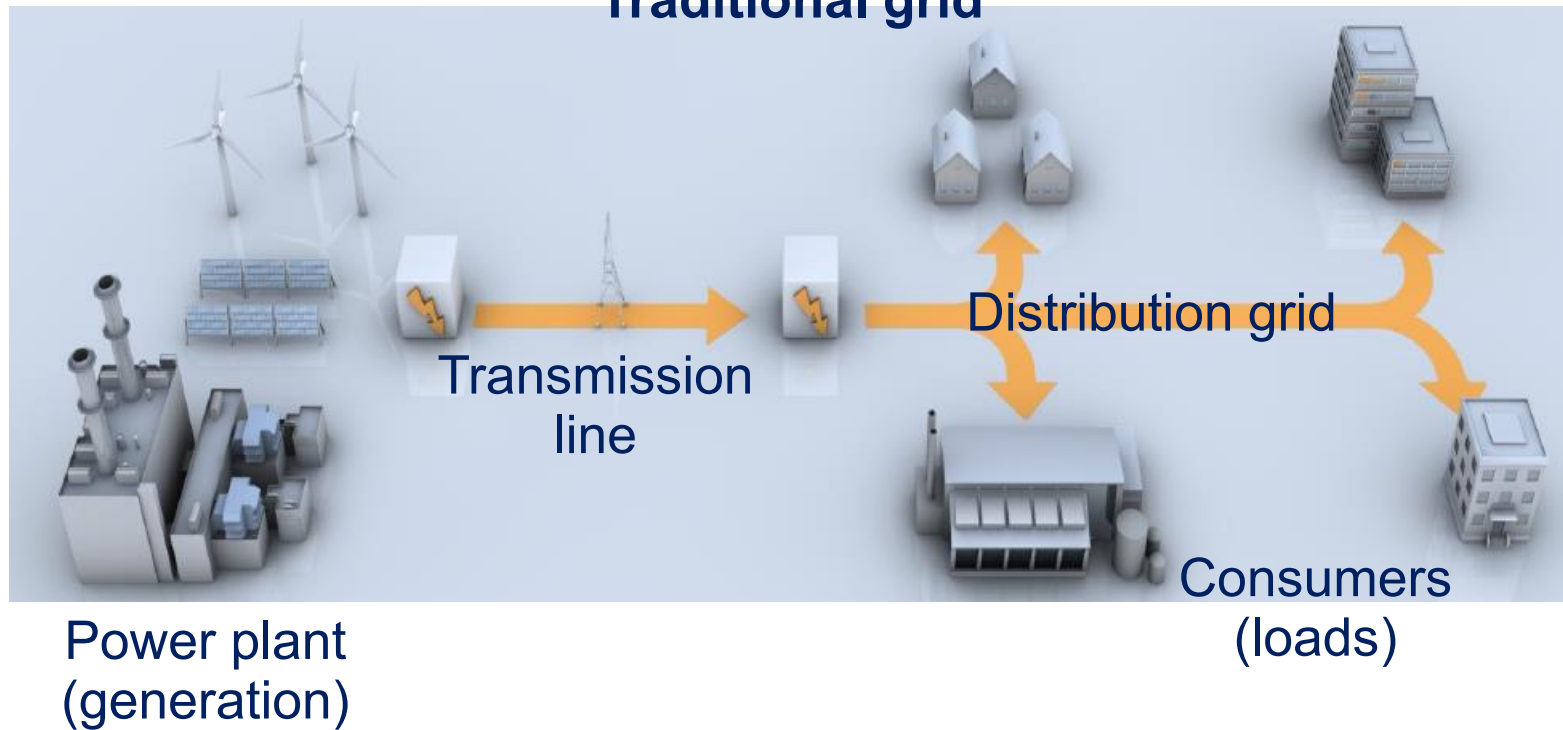
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- **Natural gas treatment plants:** interaction of separators, heat exchangers, chillers, mixers, separation columns, ...
- **Electric grids**
- **Fleets of underwater autonomous vehicles**
- **Irrigation channels**



Examples

Electric grid Traditional grid

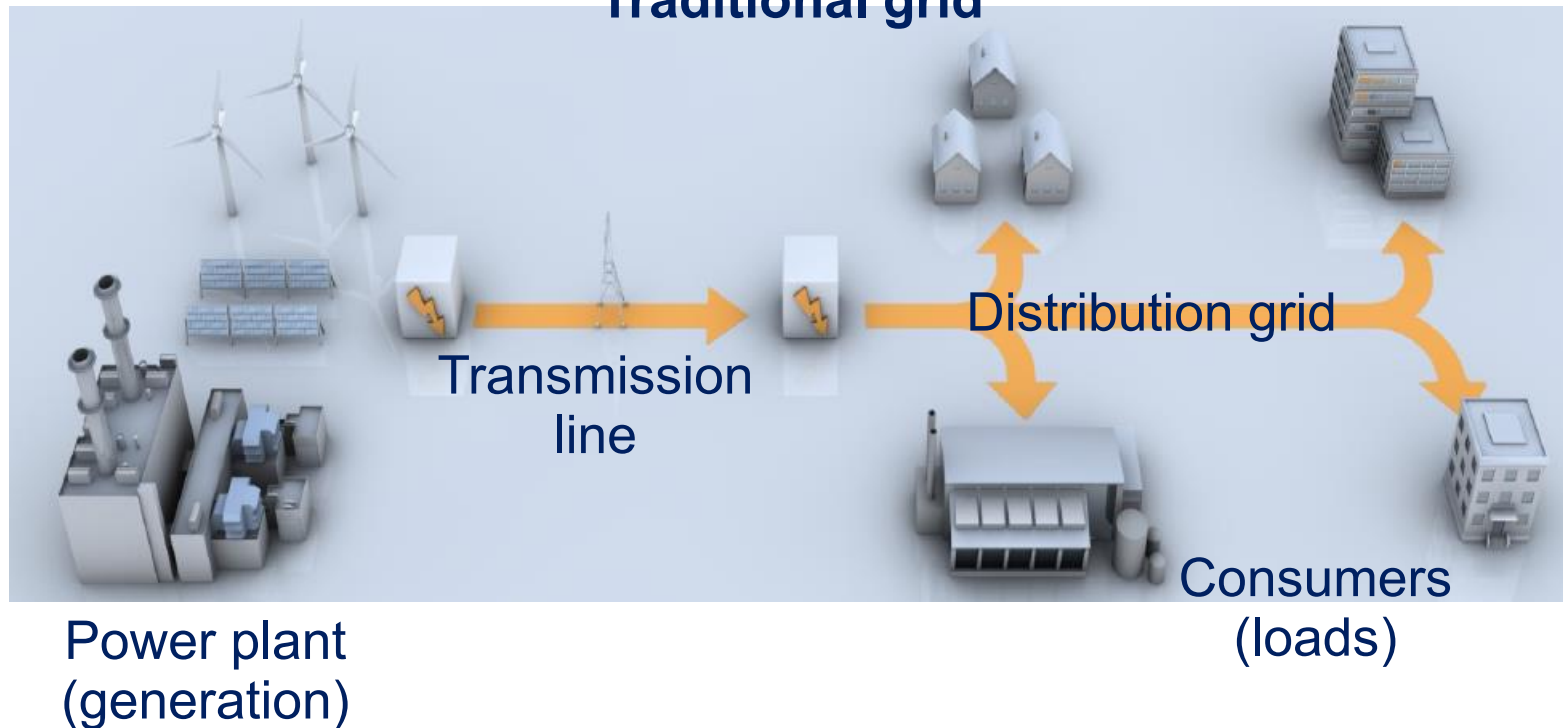


Drawbacks:

- Few generation plants far from loads
- Large use of fossil fuels

Examples

Electric grid Traditional grid



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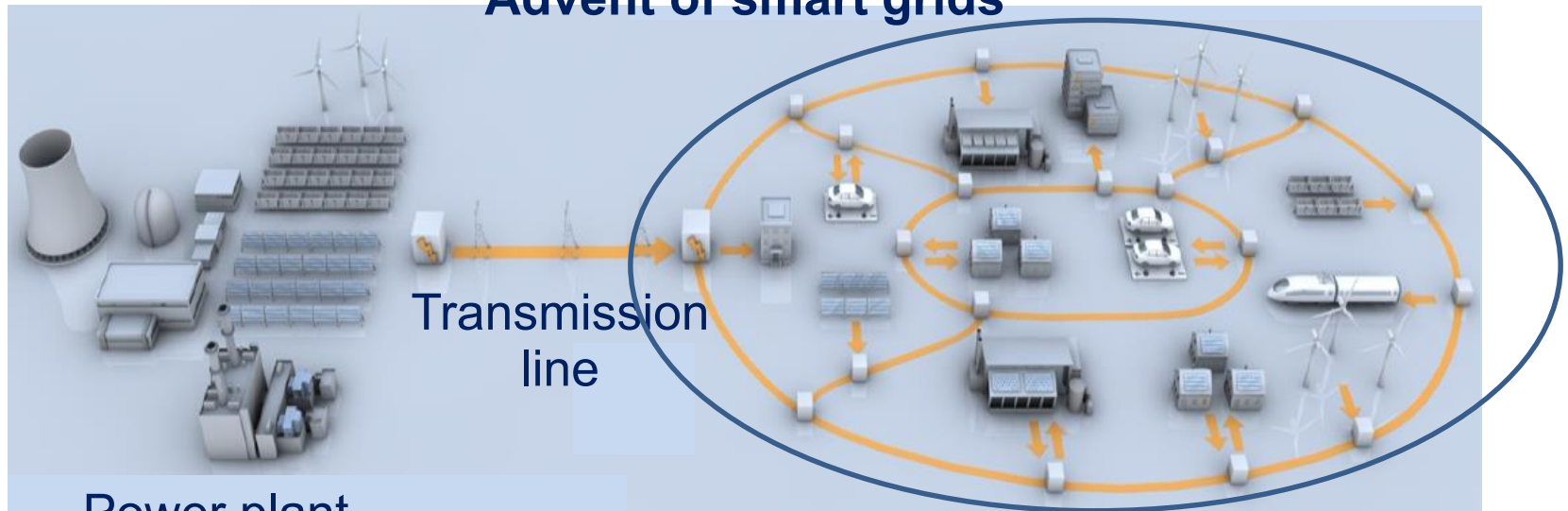
- Few generation plants far from loads
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Increasing demand and availability of local (i.e., at consumer level) and «clean» electricity (e.g., PV panels, wind turbines)!

Examples

Electric grid

Advent of smart grids



Power plant
(main
generation)

Micro-grid
(generation+
loads +
batteries)

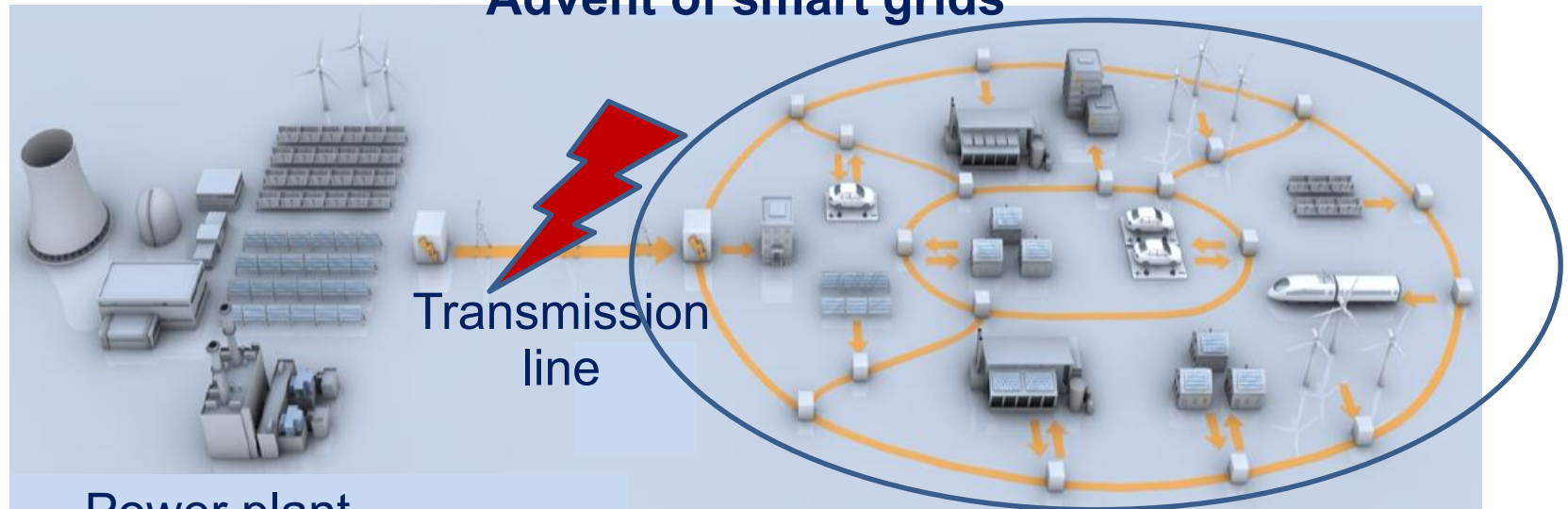
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- Distributed generation close to loads
- Increasing use of renewables
- Redundancy
- Modularity

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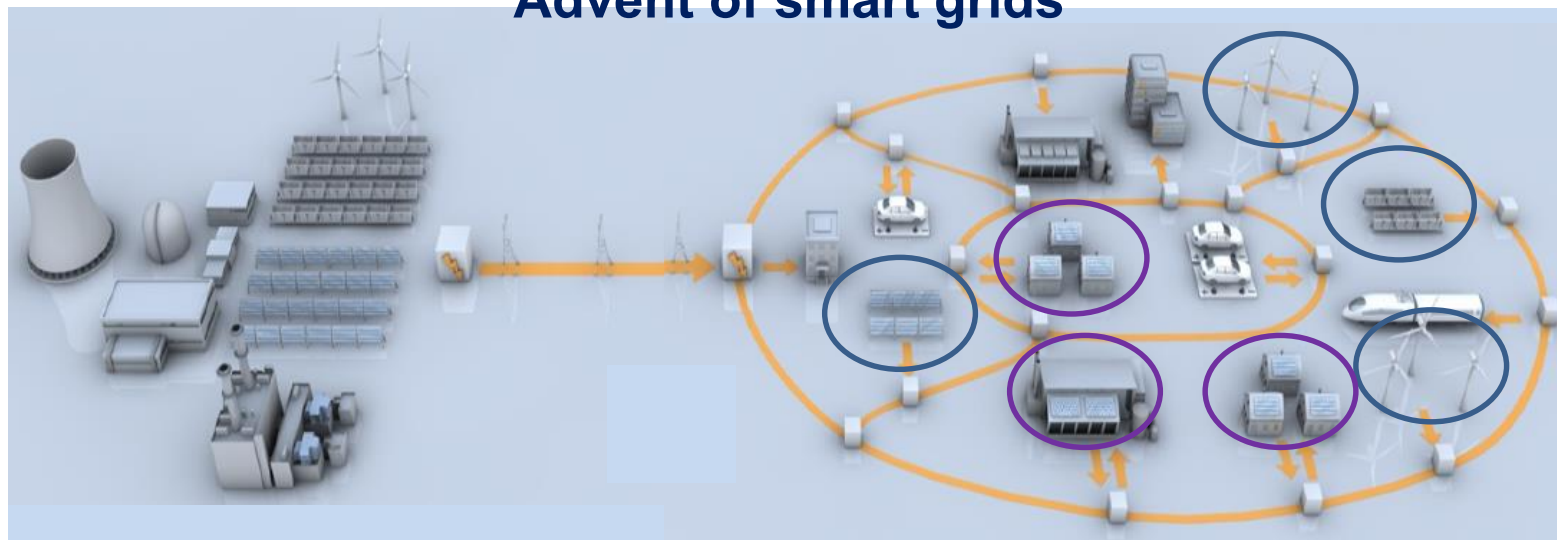
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- Possibility of operation in islanded mode (self-sufficient network of distributed generation units)

Examples

Electric grid

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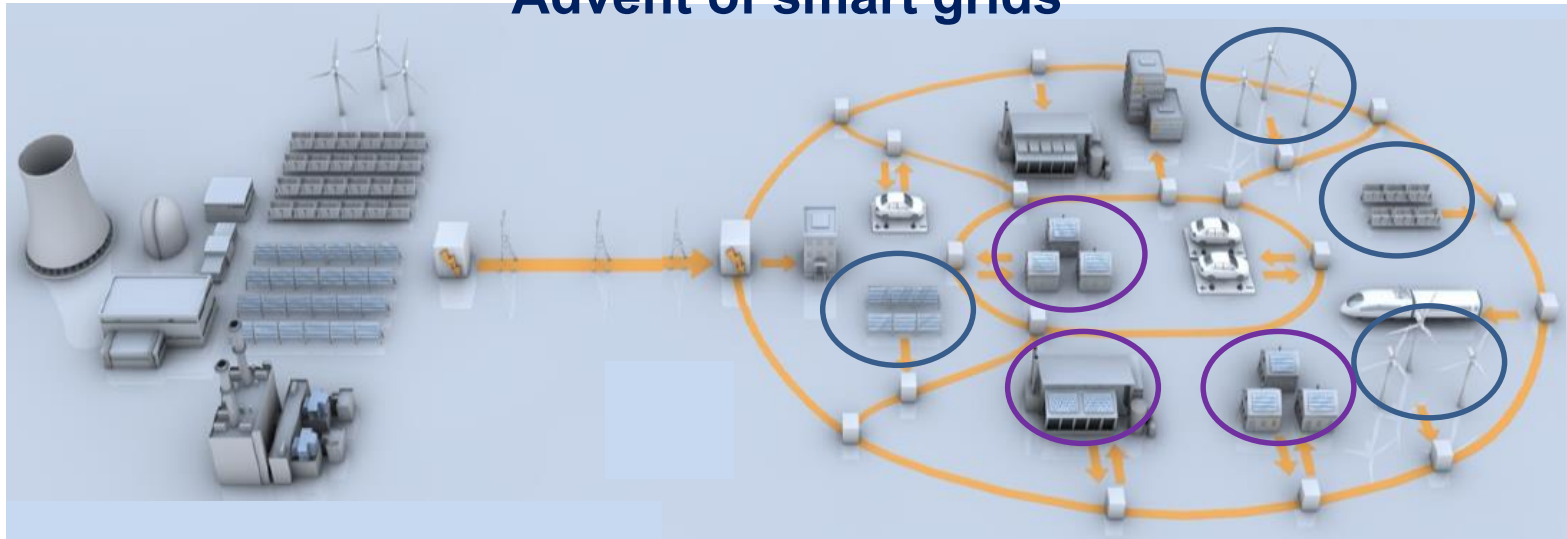


Control and optimization issues at different levels:

Examples

Electric grid

Advent of smart grids



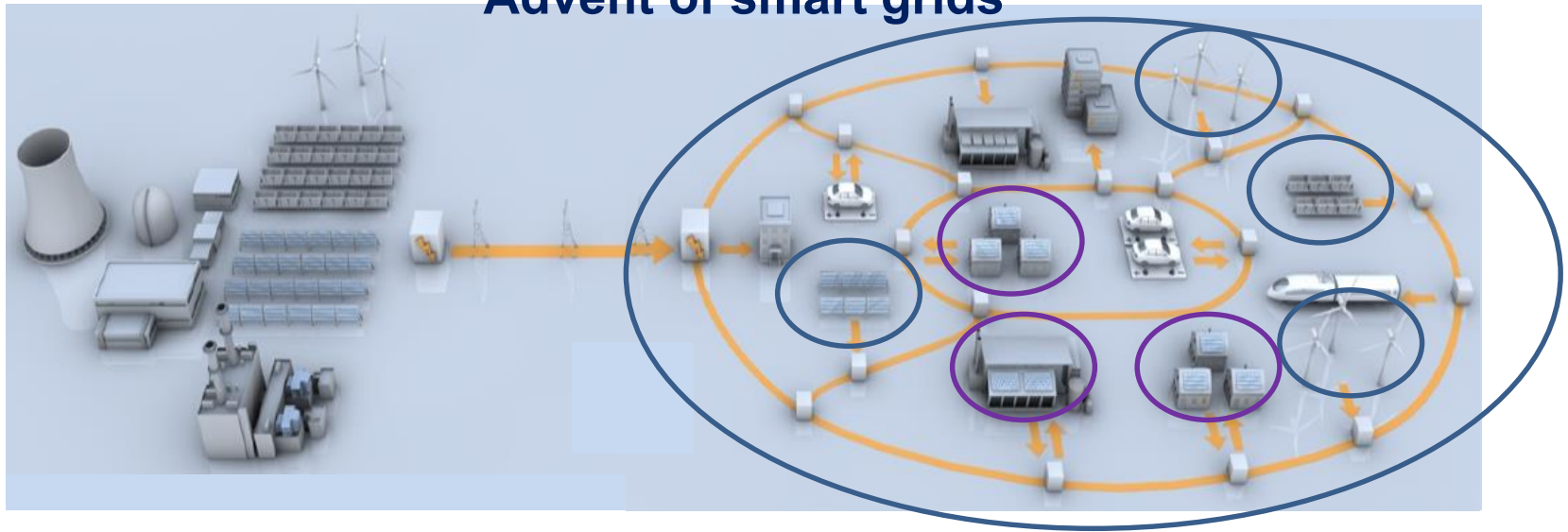
Control and optimization issues at different levels:

- Single **generation unit** level:
 - Stable operation.
 - Reactive power control .

Examples

Electric grid

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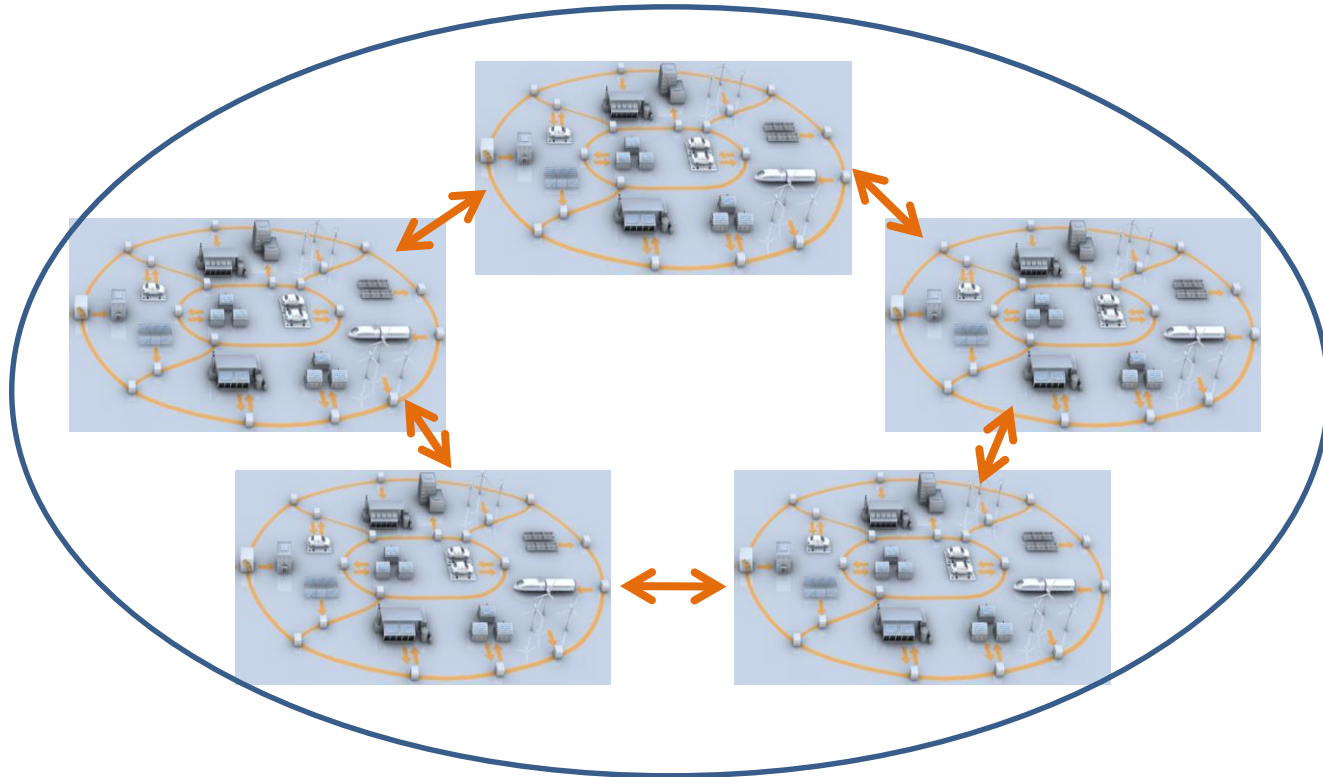


Control and optimization issues at different levels:

- Single **generation unit** level:
 - Stable operation.
 - Reactive power control .
- Single **microgrid** level:
 - Guarantee the fulfillment of energy needs (with possibly controllable loads).
 - Maximize gain from local production.
 - Minimize purchase from the grid.
 - Frequency/voltage regulation (especially in case of islanded operation).

Examples

Electric grid



- **Aggregator level:**
 - Cluster of microgrids that take the role of a **(virtual) power plant**
 - Active power reserves
 - Reactive power reserves

Examples

Electric grid

Traditional centralized control structures are not suitable

Need of more complex control architectures

- Control at different levels
- Coordination

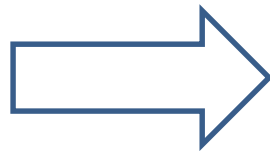
Examples

Electric grid

Traditional centralized control structures are not suitable

Need of more complex control architectures

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Novel hierarchical **and**
distributed control architectures

Examples

Fleets of underwater autonomous vehicles for sensing

Goal: **design a** mobile sensor **network** to collect measurements from a field

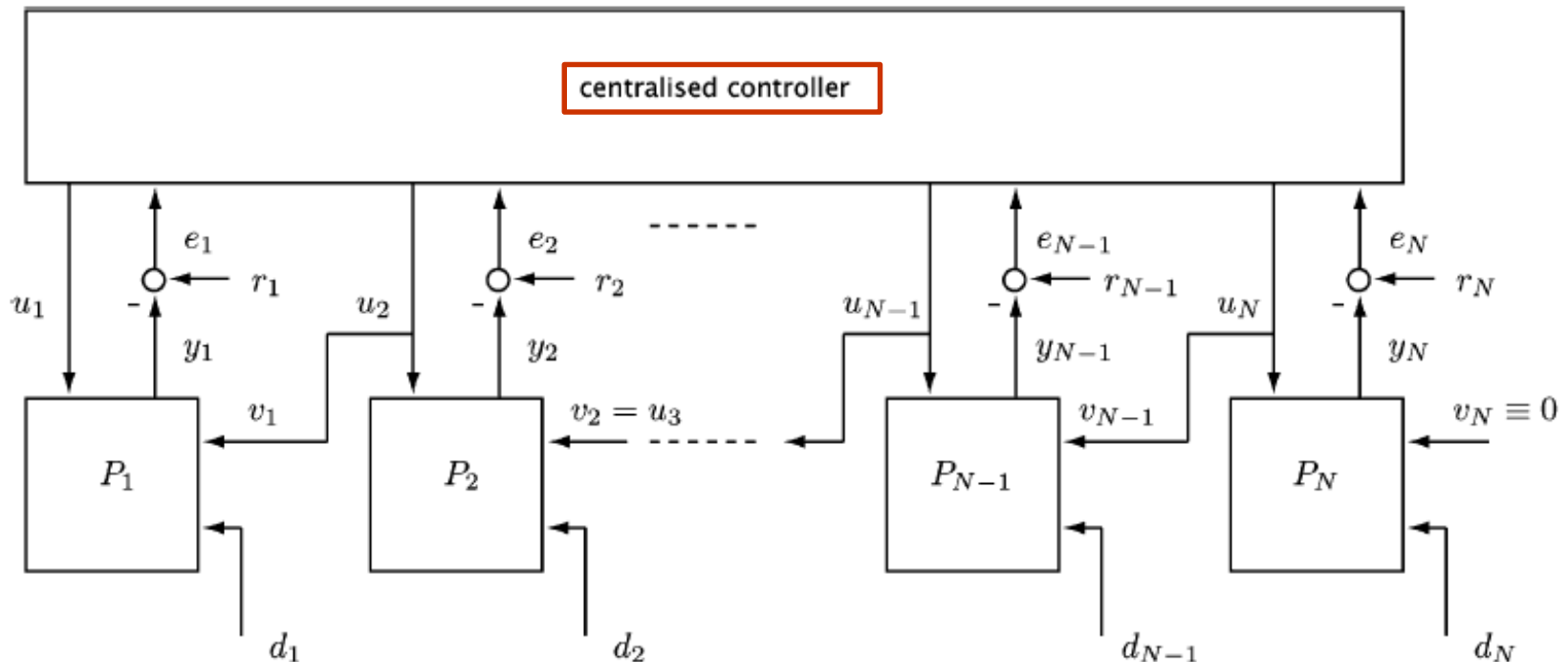
- e.g., salinity, temperature, chlorophyll fluorescence (i.e., a proxy for the concentration of phytoplankton, bottom of marine food chain, indicator of the state of the marine ecosystem and of the atmosphere)
- optimal measurements: sensors should **move in a rigid formation**.

From: N. E. Leonard *et al.* Collective motion, sensor networks, and ocean sampling. *Proceedings of the IEEE*, 95 (1), pp. 48-74

Examples

Irrigation networks

- **Centralized** computation
 - **Centralized** information structure
- Not compatible with a possibly very large-scale system!



Examples

Irrigation networks

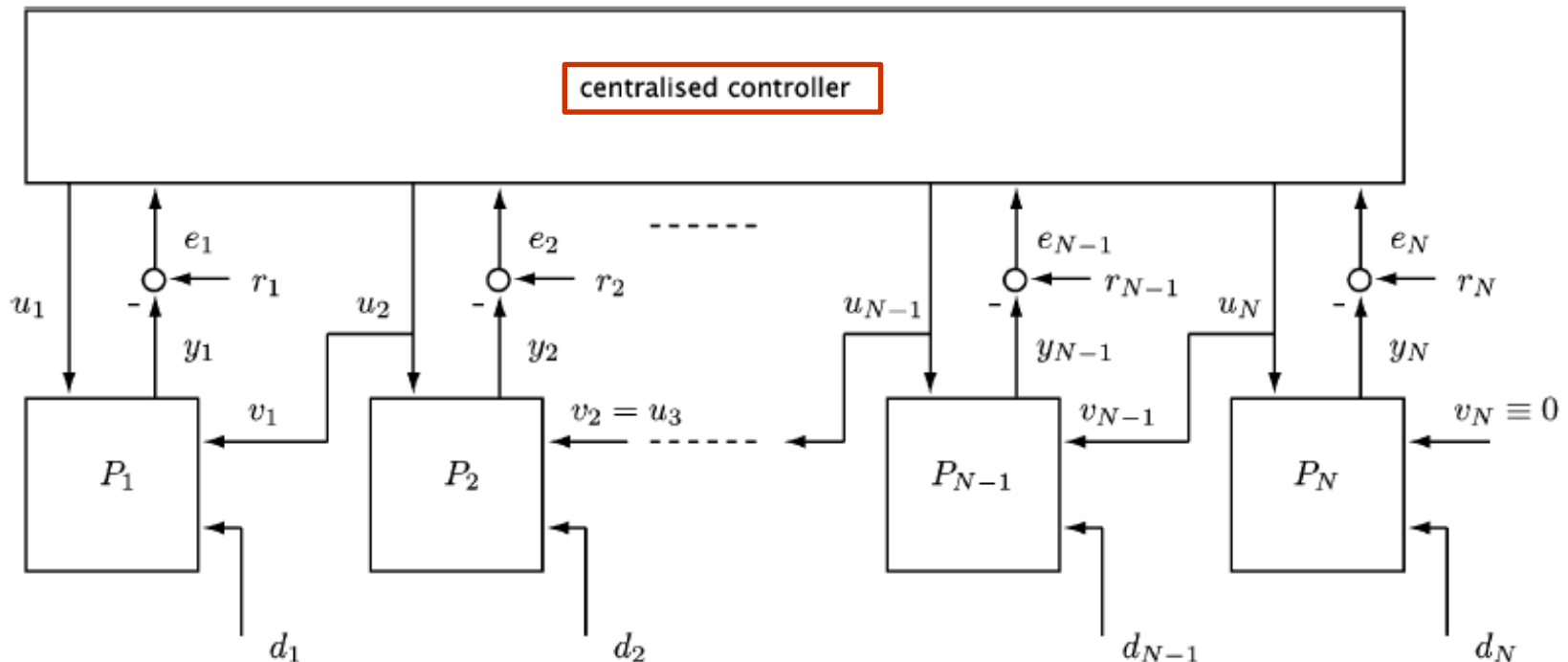
- **Centralized** computation
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Not compatible with a possibly very large-scale system!



Distributed or decentralized solutions are necessary!

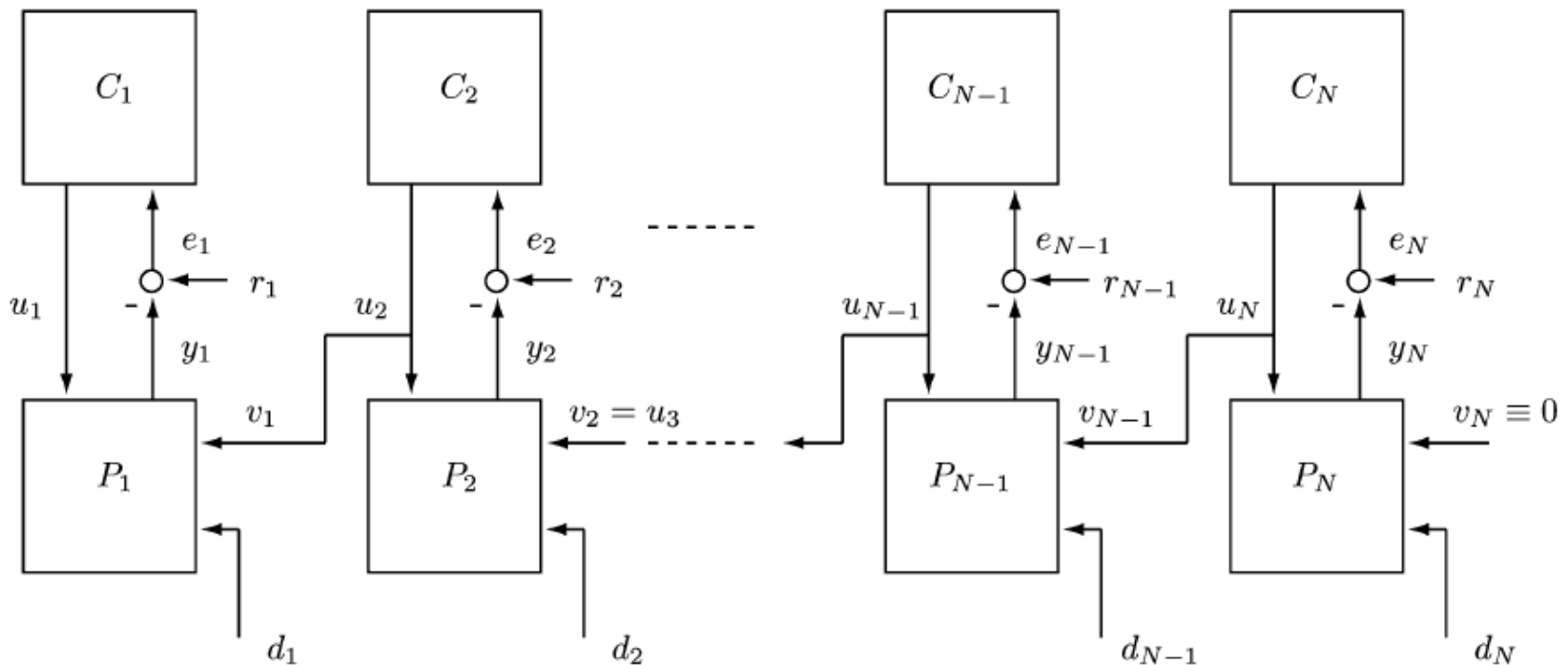


Examples

Irrigation networks

- Local computation
- **Decentralized** information structure

Actions, taken at local pool levels, are computed locally based on local information

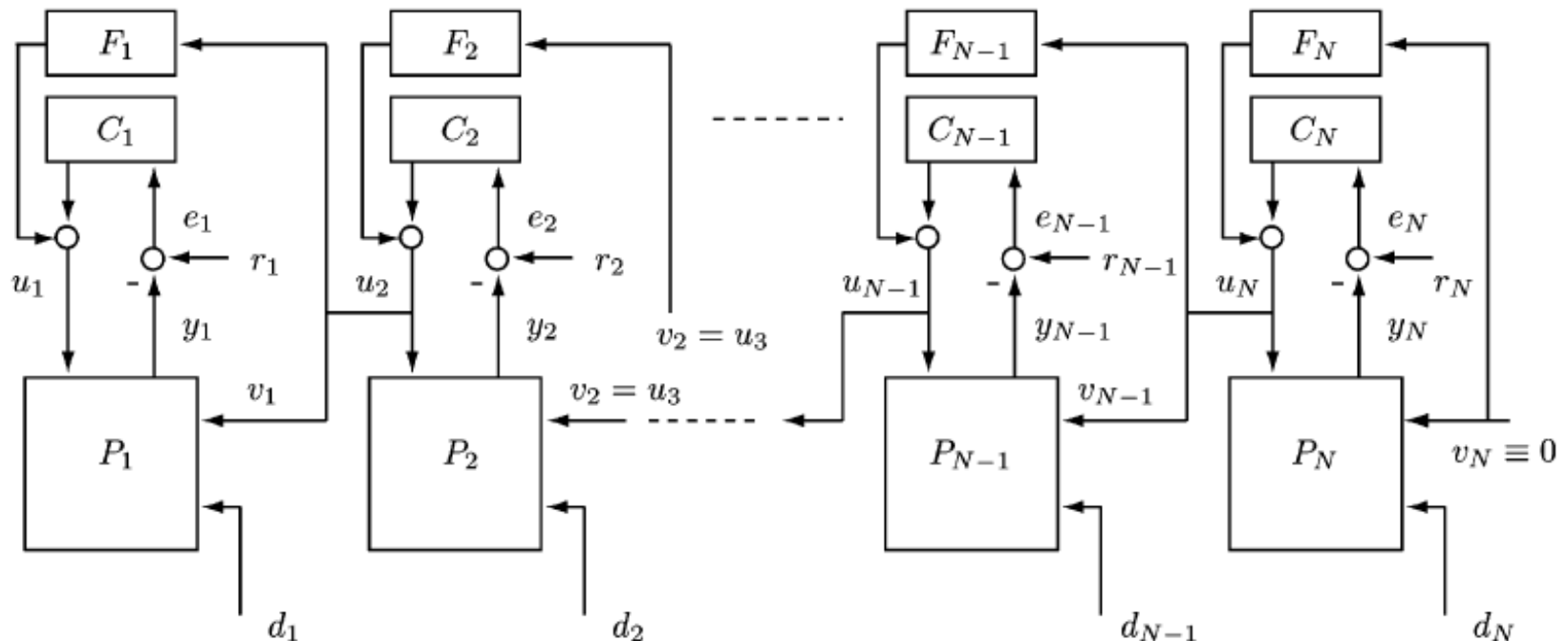


Examples

Irrigation networks

Actions, taken at local pool levels, are computed locally based on local information + information transmitted by neighboring gates' control systems

- Local computation
- **Distributed** information structure



Large-scale and complex systems

All the mentioned examples (and many more...) are systems characterized by a (complex) network of many heterogeneous systems. They are referred to, in the literature, as

- **Complex systems**
- **Large-scale systems**

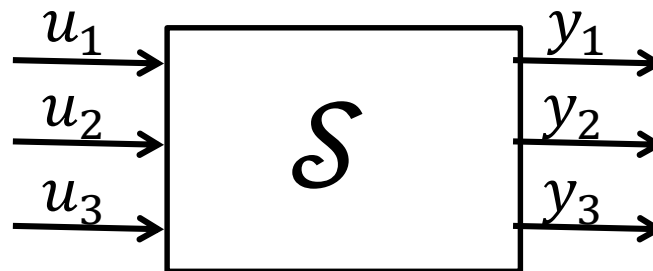
«**Multi-agent systems**» is a subclass of network-structured systems, characterized by the fact that the dynamics of the subsystems are independent.

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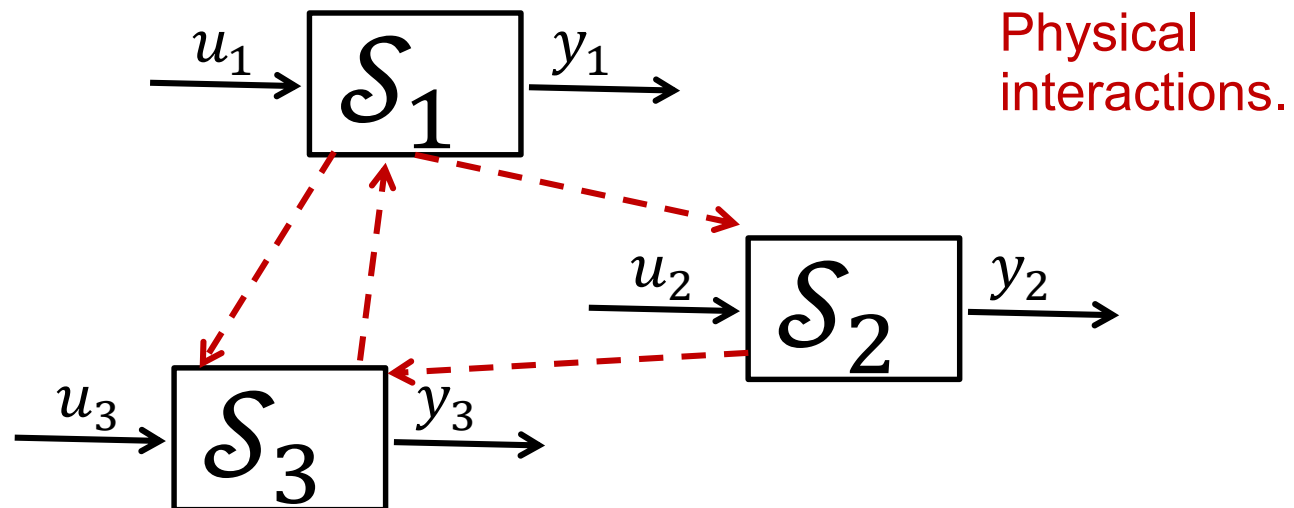


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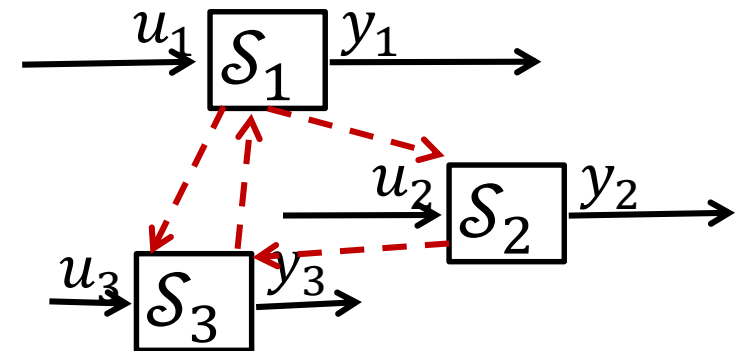
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Large-scale and complex systems

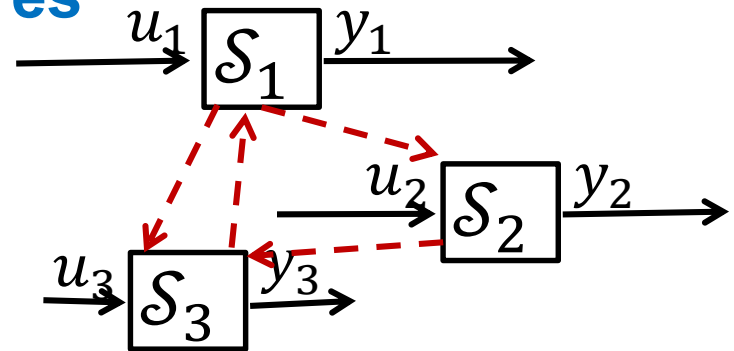
Challenges

- **High dimensionality:** large number of
 - subsystems in which the overall system can be subdivided;
 - variables characterizing the system state;
 - outputs;
 - variables to be manipulated for control.
- **Information structure constraints:** in view of the system sparsity, physical constraints may prevent information to be transmissible without limitations.
- **Uncertainties:** they may affect the single subsystems and their interconnections (which may also be time varying).



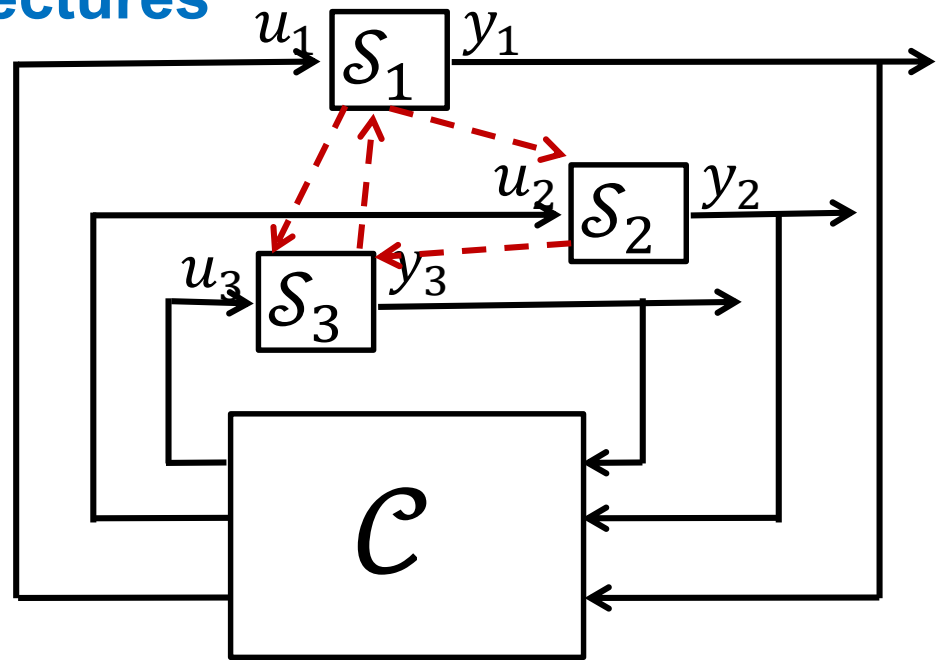
Control schemes and architectures

Centralized control structures:



Control schemes and architectures

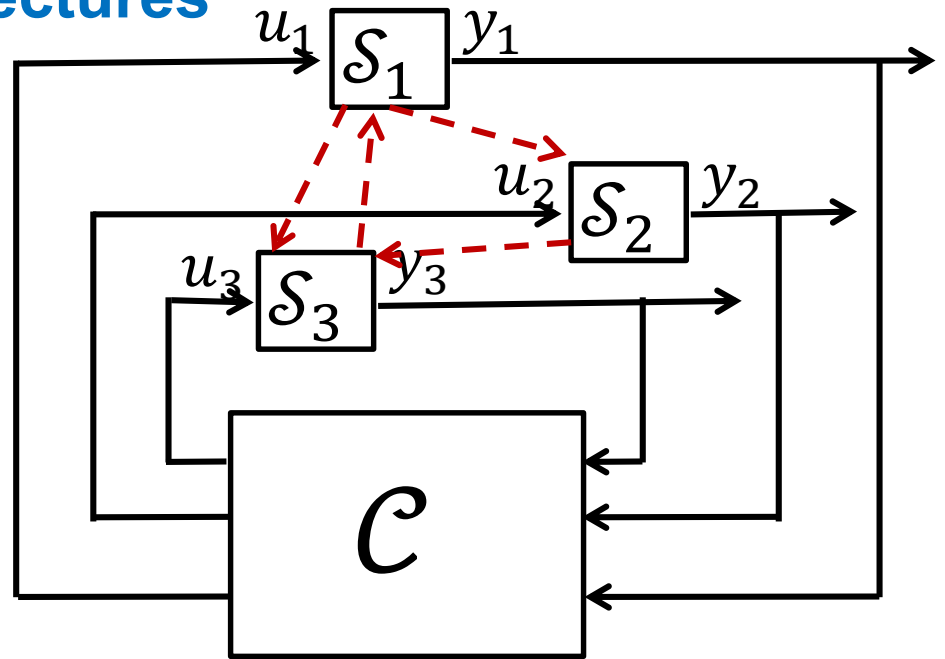
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Control schemes and architectures

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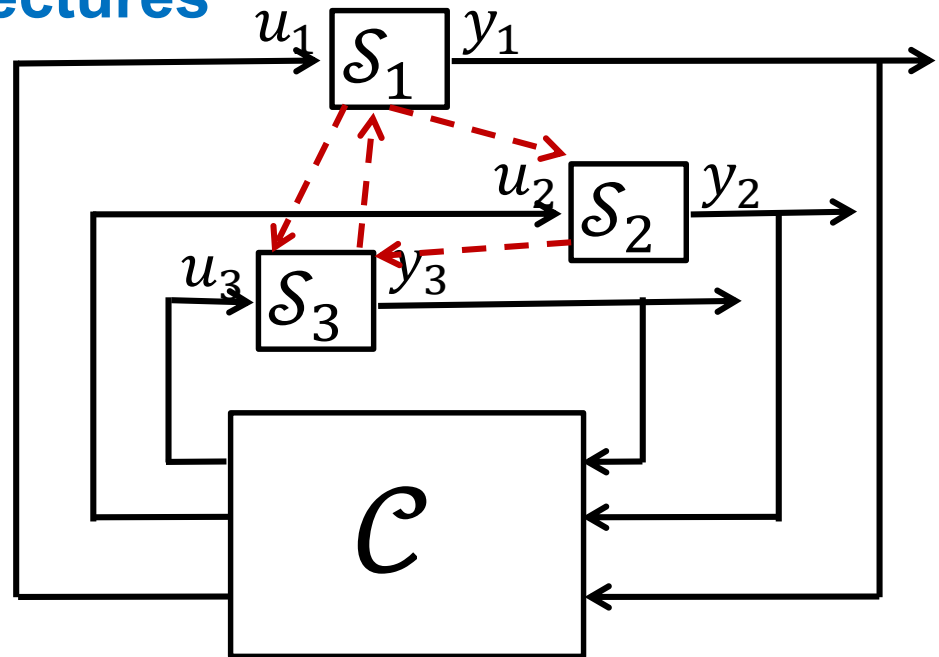
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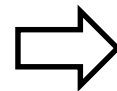
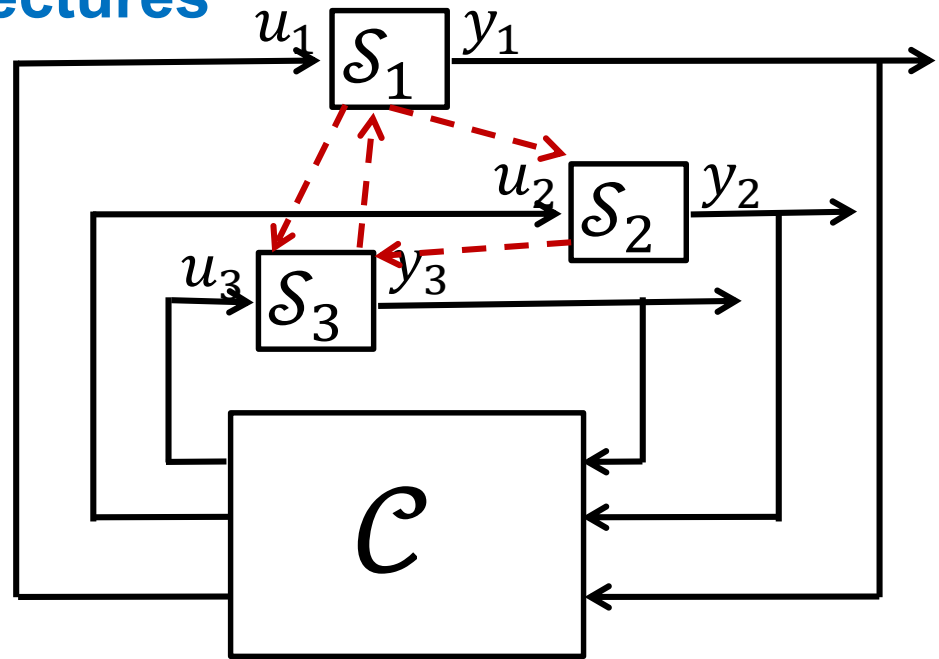


- Possibly large distances
- Large number of links

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-> Downtime!



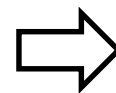
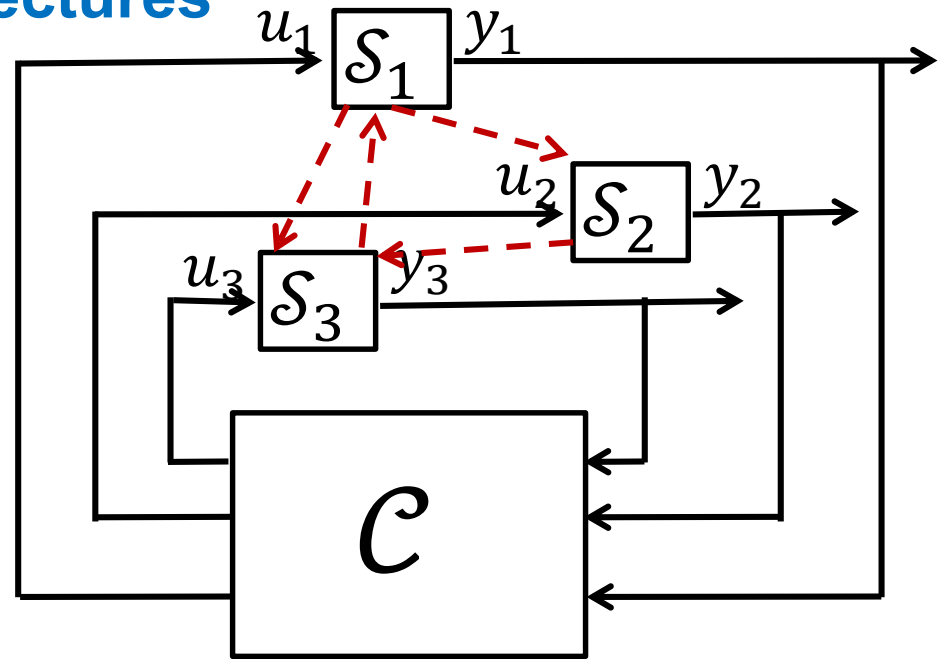
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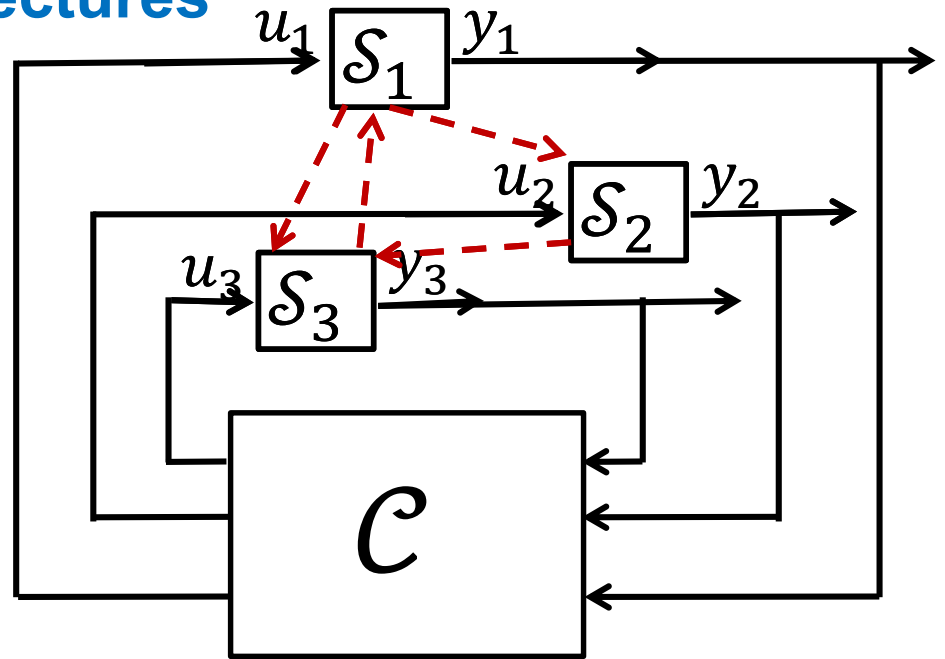
⇒ **They may not be feasible!**



- Possibly large distances
- Large number of links
- For computational reasons
- For communication reasons
- Because they lack the necessary flexibility

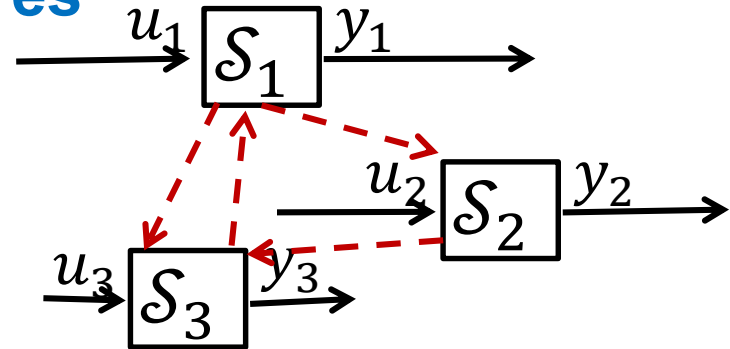
Control schemes and architectures

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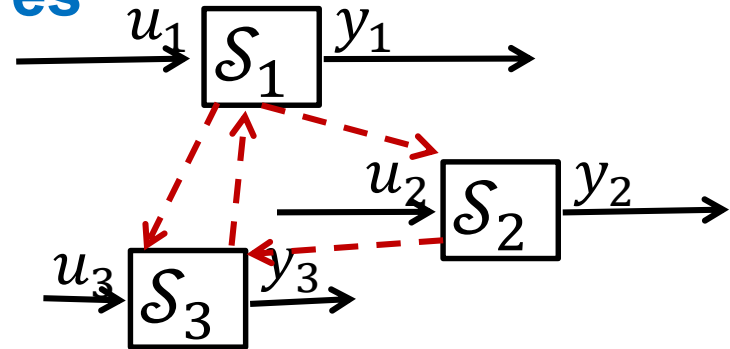
If large-scale/complex systems must be controlled, their analysis and synthesis become very complex.

That is, these tasks cannot be solved just by using faster computers and larger memories.

They necessitate new ideas for decomposing and dividing the analysis and control problems of the overall system into rather independent subproblems and for dealing with the uncertainties and with the lack of information about the system to be controlled.

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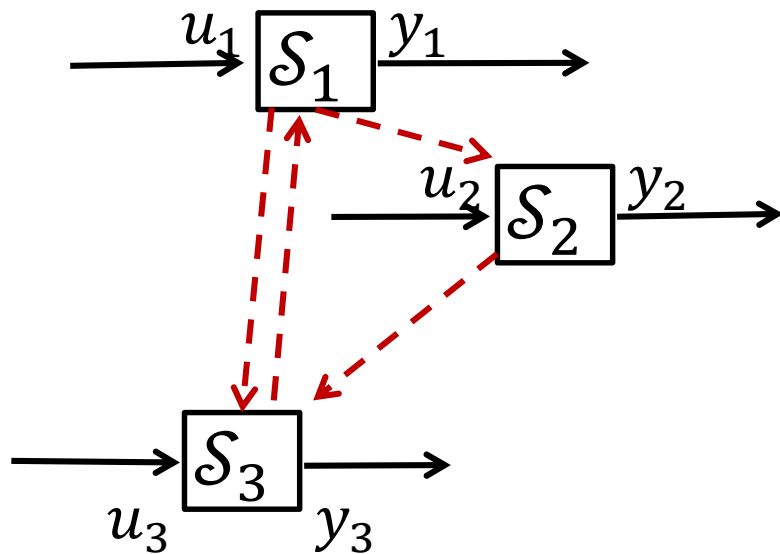
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The definition of «largeness» of a system is itself subjective, taking a pragmatic viewpoint: a system is considered to be large-scale whenever it is necessary to partition it into a number of interconnected units either for conceptual or for computational reasons.

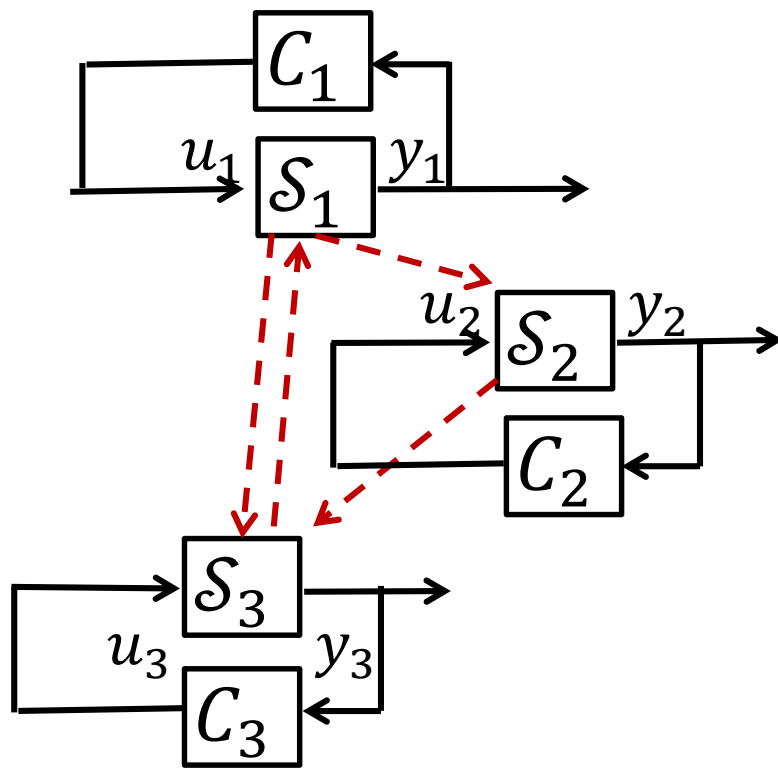
Control schemes and architectures

«**Decomposition-based**» approach: decompose the control problem into subproblems and make each subsystem controlled by a local feedback loop.



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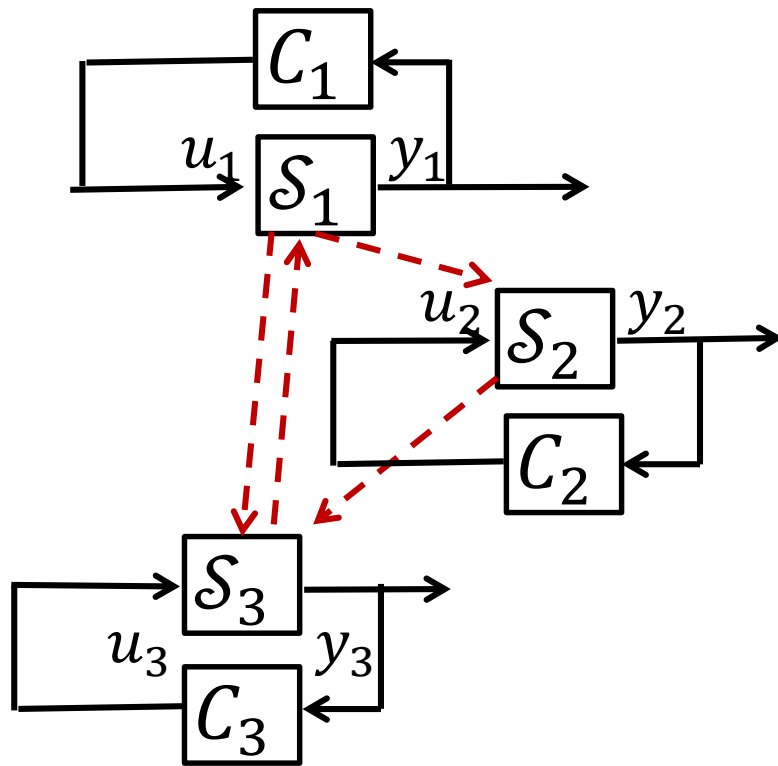


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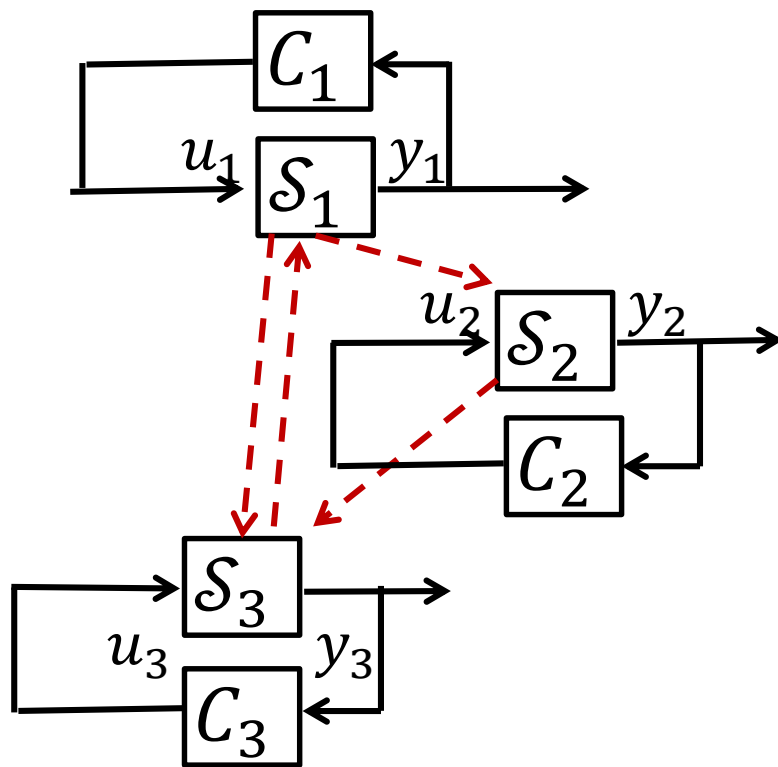
- **Reduced computational load** in control action computation.

- Fast computation.
- **Scalability**, i.e., the size of the reduced problems is limited, regardless of the size of the large-scale one.



Control schemes and architectures

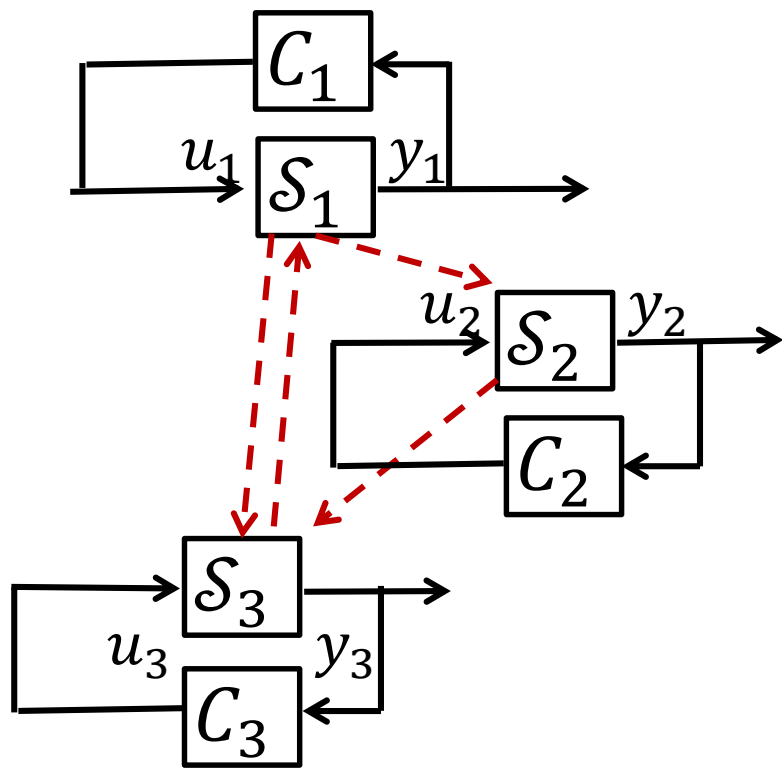
«**Decomposition-based**» approach: decompose the control problem into subproblems and make each subsystem controlled by a local feedback loop.



- **Reduced computational load** in control action computation.
- **Reduced communication burden** (wired):
 - Local transducer → local control station
 - Local control station → local actuator
- Fast computation.
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- No (or limited) transmission delays
- No (or limited) transmission failures

Control schemes and architectures

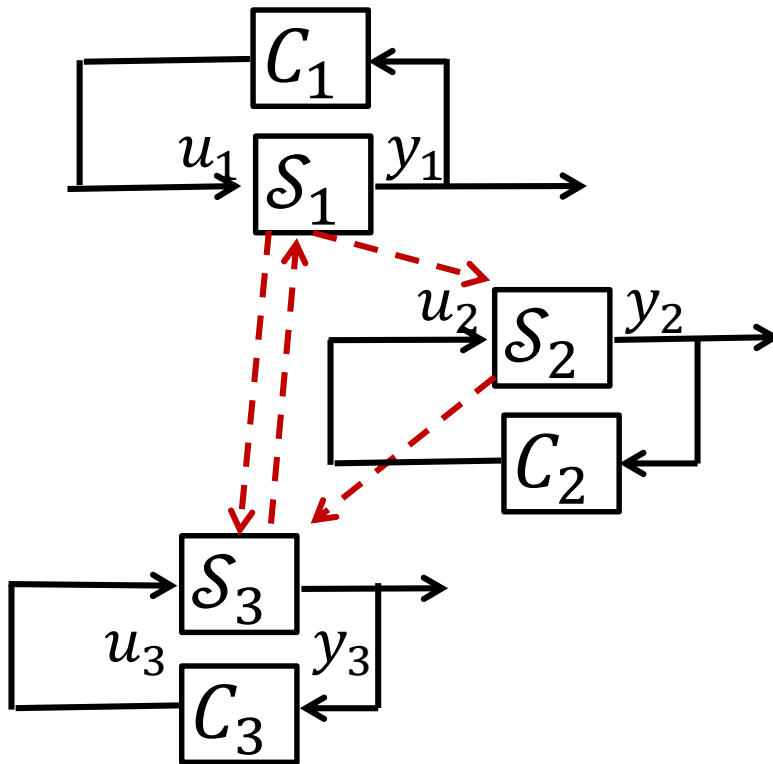
«**Decomposition-based**» approach: decompose the control problem into subproblems and make each subsystem controlled by a local feedback loop.



- **Reduced computational load** in control action computation.
- **Reduced communication burden** (wired):
 - Local transducer → local control station
 - Local control station → local actuator
- **Reduced memory** required:
 - Store local subsystem model
 - Store local variables
- Local **model uncertainties can be reduced**
- Local controllers can be made **robust** with respect to uncertain interconnections
- Fast computation.
- **Scalability**, i.e., the size of the reduced problems is limited, regardless of the size of the large-scale one.
- No (or limited) transmission delays
- No (or limited) transmission failures
- Changes in a part does not require the re-design of the overall system (**modularity**).

Control schemes and architectures

Decentralized control

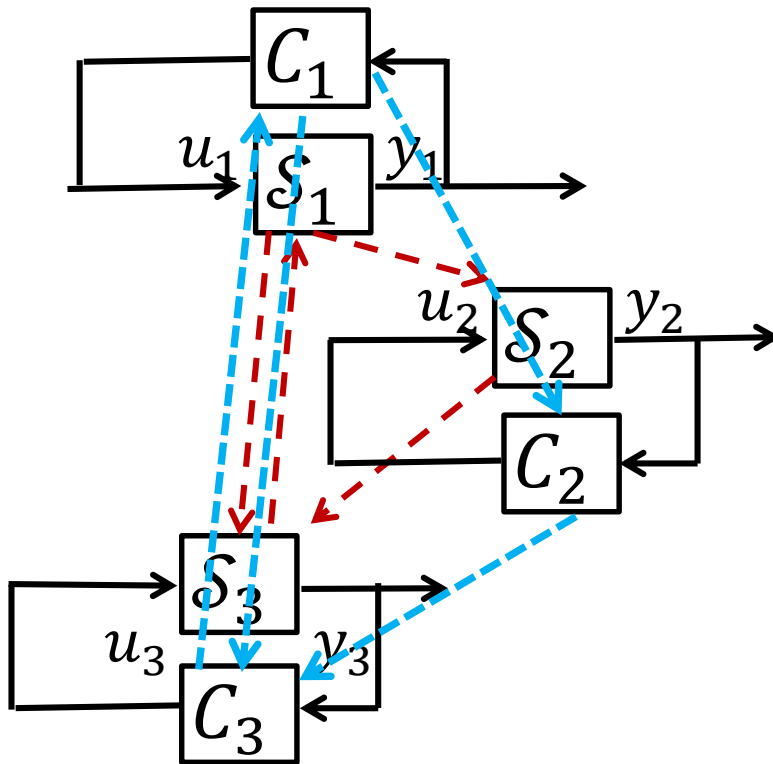


- Local controllers attached to each input/output channel
- Absence of communication between controllers limits the achievable performance

Control schemes and architectures

Distributed control

- Local controllers attached to each input/output channel (as in the decentralized scheme)
- Controllers can communicate: one can tune **the trade-off between communication burden and performance**



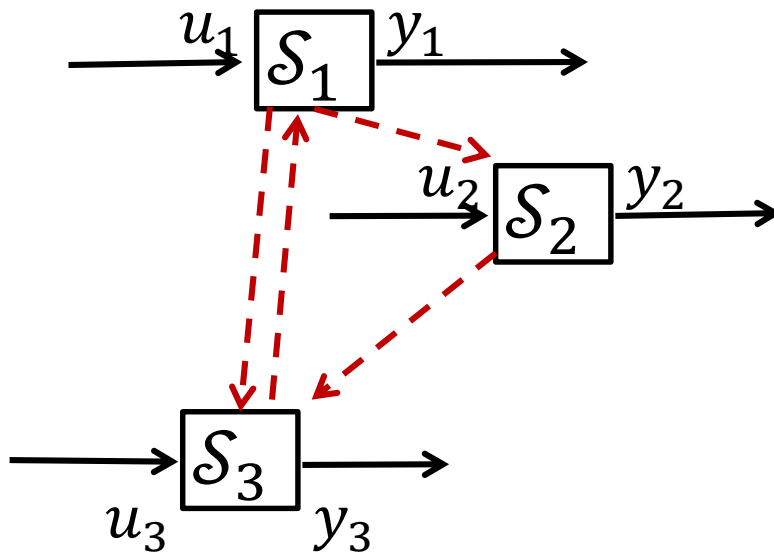
- Middle ground between centralized and decentralized schemes.
- Communication graph is often part of the design problem.
- **Challenges due to network non idealities (delays, packet dropouts, etc.)...**

Control schemes and architectures

Three different “network” layers

Layer 1: **system interconnections network**

- Dynamic interactions
- Coupling constraints
- Common goal



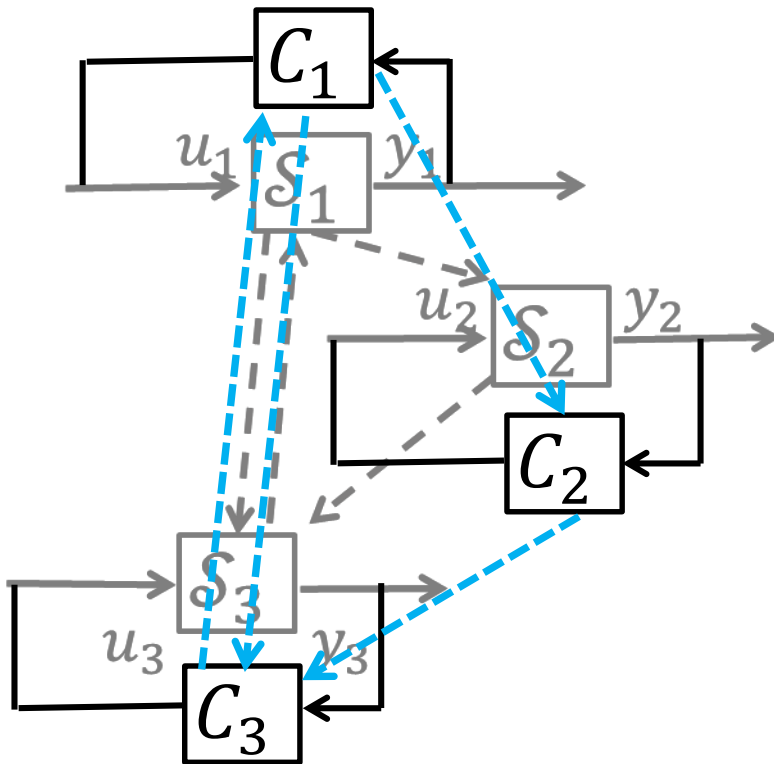
Subsystems whose dynamics and/or objective functions have a neighborhood structure that can be described by a **graph**.

Control schemes and architectures

Three different “network” layers

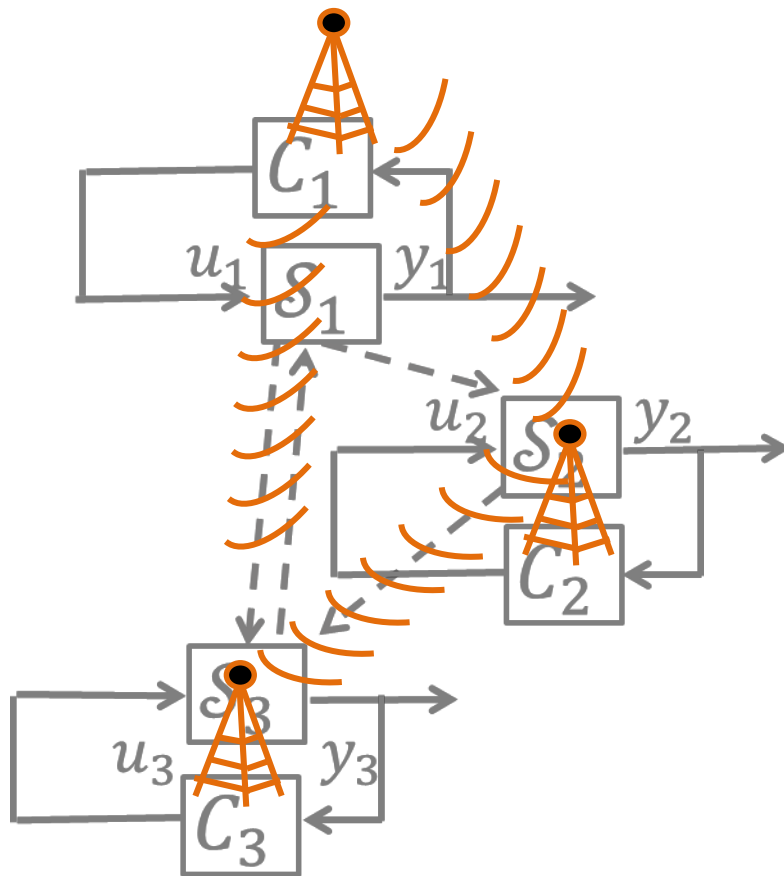
Layer 2: **decision (control) network**

- The decision taken by an agent may depend upon the decision taken (possibly at different time instants) by other agents.
- The decision network may not overlap with the system interconnection (e.g., in decentralized control the decision network is absent)



Control schemes and architectures

Three different “network” layers



Layer 3: **communication network** (infrastructure)

- How data are transmitted from one agent to the others?
- The decision network may not overlap with the communication network (multi-hop communication is possible).

Challenges and issues

Issues addressed in the following lectures:

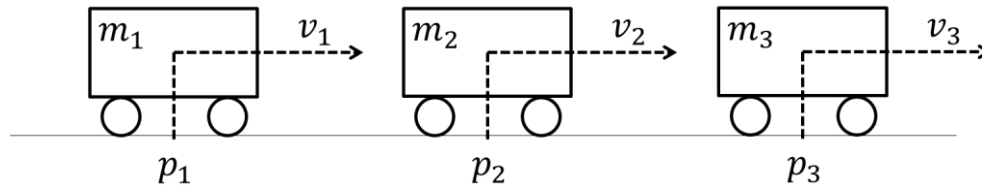
- Is it (structurally) possible to design a decentralized/distributed controller that confers stability?

Challenges and issues

Issues addressed in the following lectures:

- Is it (structurally) possible to design a decentralized/distributed controller that confers stability?
- How to design a decentralized/distributed controller?

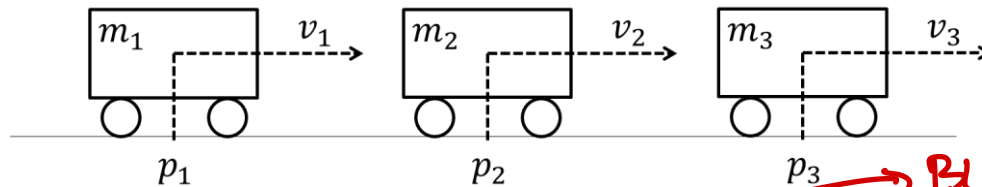
Examples: DETACHED CARTS ON A RAILWAY



$$m_i \ddot{p}_i = m_i \dot{v}_i = -h_i v_i + u_i$$

$$\begin{bmatrix} \dot{p}_i \\ \dot{v}_i \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & -\frac{h_i}{m_i} \end{bmatrix} \begin{bmatrix} p_i \\ v_i \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{1}{m_i} \end{bmatrix} u_i$$

Examples: DETACHED CARTS ON A RAILWAY



DYNAMICS
DECOUPLED

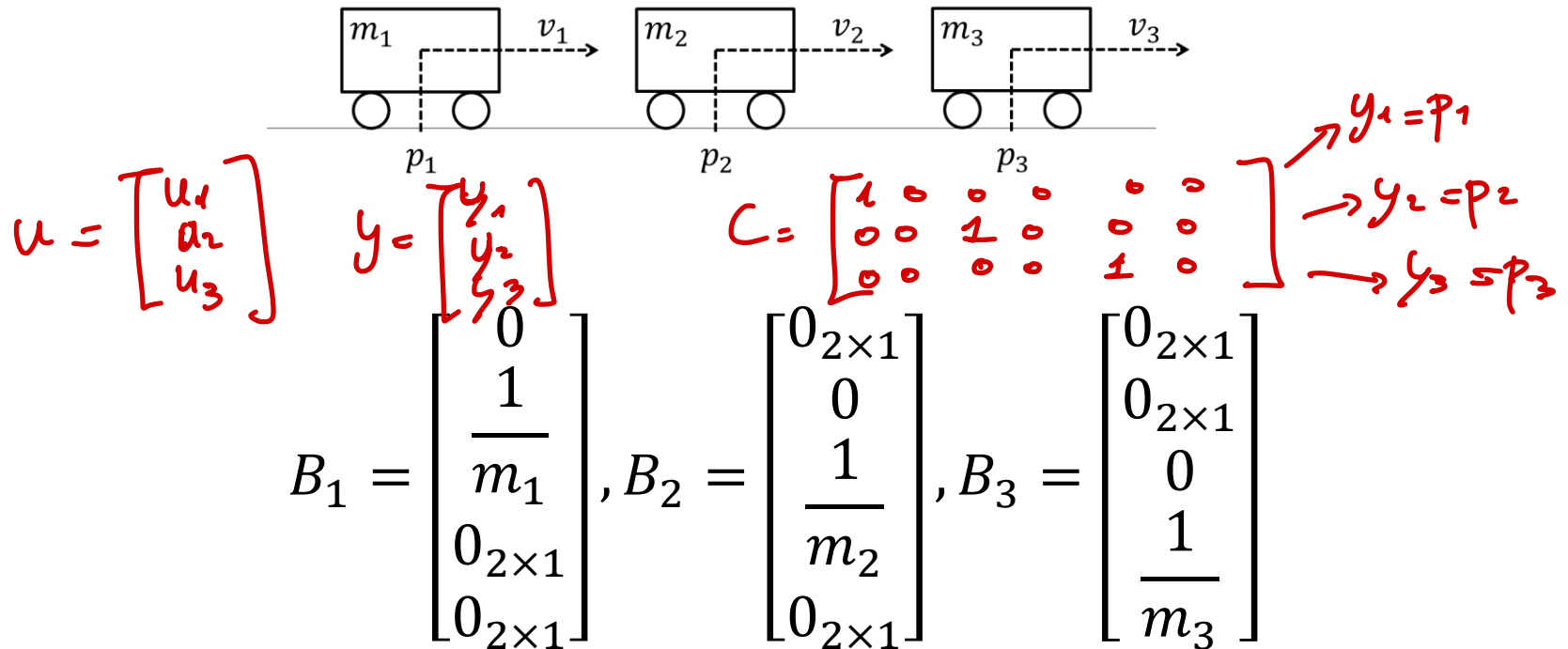
BLOCK-DIAGONAL
MATRIX

$$x = \begin{bmatrix} p_1 \\ v_1 \\ p_2 \\ v_2 \\ p_3 \\ v_3 \end{bmatrix}, A = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & -\frac{h_1}{m_1} & 0_{2 \times 2} & 0_{2 \times 2} & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0_{2 \times 2} & 0 & 0 & -\frac{h_2}{m_2} & 0_{2 \times 2} & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0_{2 \times 2} & 0_{2 \times 2} & 0 & 0 & 0 & -\frac{h_3}{m_3} \end{bmatrix}, B = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ \frac{1}{m_1} & 0_{2 \times 1} & 0_{2 \times 1} & 0_{2 \times 1} & 0_{2 \times 1} & 0_{2 \times 1} \\ 0_{2 \times 1} & 0 & 1 & 0 & 0 & 0 \\ 0_{2 \times 1} & 0_{2 \times 1} & 0_{2 \times 1} & \frac{1}{m_2} & 0_{2 \times 1} & 0_{2 \times 1} \\ 0_{2 \times 1} & 0_{2 \times 1} & 0_{2 \times 1} & 0_{2 \times 1} & 1 & 0 \\ 0_{2 \times 1} & 0_{2 \times 1} & 0_{2 \times 1} & 0_{2 \times 1} & 0_{2 \times 1} & \frac{1}{m_3} \end{bmatrix}, u = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix}$$

INPUT
DECOUPLED
SUBSYSTEMS

$$C = \begin{bmatrix} 1 & 0 & 0_{1 \times 2} & 0_{1 \times 2} \\ 0_{1 \times 2} & 1 & 0 & 0_{1 \times 2} \\ 0_{1 \times 2} & 0_{1 \times 2} & 1 & 0 \end{bmatrix}, y = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$$

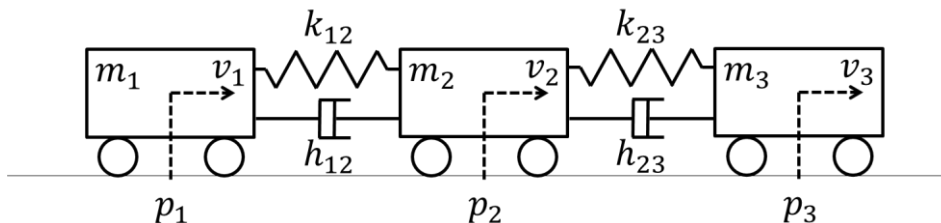
Examples: DETACHED CARTS ON A RAILWAY



$$C_1 = \begin{bmatrix} 1 & 0 & 0_{1 \times 2} & 0_{1 \times 2} \end{bmatrix}, C_2 = \begin{bmatrix} 0_{1 \times 2} & 1 & 0 & 0_{1 \times 2} \end{bmatrix},$$

$$C_3 = \begin{bmatrix} 0_{1 \times 2} & 0_{1 \times 2} & 1 & 0 \end{bmatrix}$$

Examples: CARTS ON A RAILWAY CONNECTED THROUGH SPRINGS AND DAMPERS



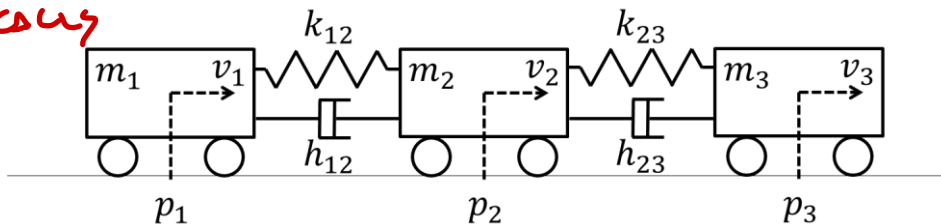
$$m_1 \ddot{p}_1 = m_1 \dot{v}_1 = -h_1 v_1 + u_1 - k_{12}(p_1 - p_2) - h_{12}(v_1 - v_2)$$

$$m_2 \ddot{p}_2 = m_2 \dot{v}_2 = -h_2 v_2 + u_2 - k_{12}(p_2 - p_1) - h_{12}(v_2 - v_1) \\ - k_{23}(p_2 - p_3) - h_{23}(v_2 - v_3)$$

$$m_3 \ddot{p}_3 = m_3 \dot{v}_3 = -h_3 v_3 + u_3 - k_{23}(p_3 - p_2) - h_{23}(v_3 - v_2)$$

Examples: CARS ON A RAILWAY CONNECTED THROUGH SPRINGS AND DAMPERS

SYSTEM'S DYNAMICS
COUPLED



BANDIED
MATRIX

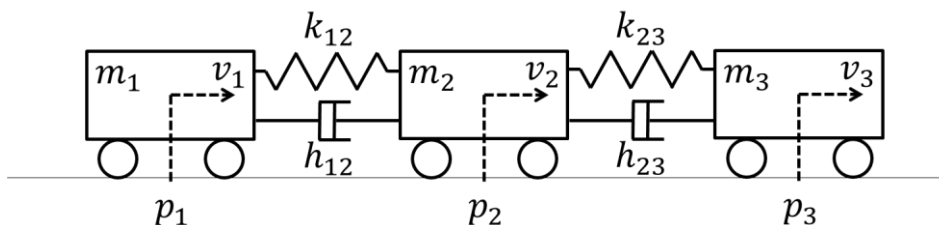
$$x = \begin{bmatrix} p_1 \\ v_1 \\ p_2 \\ v_2 \\ p_3 \\ v_3 \end{bmatrix}, A = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ -\frac{k_{12}}{m_1} & -\frac{(h_{12} + h_{12})}{m_1} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ \frac{k_{12}}{m_2} & \frac{h_{12}}{m_2} & -\frac{(k_{12} + k_{23})}{m_2} & -\frac{(h_{12} + h_{23} + h_{23})}{m_2} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ \frac{k_{23}}{m_3} & \frac{h_{23}}{m_3} & 0 & 0 & -\frac{k_{23}}{m_3} & -\frac{(h_{23} + h_{23})}{m_3} \end{bmatrix}$$

Handwritten annotations: Red circles highlight the $0_{2 \times 2}$ blocks in the top-right and bottom-right of the matrix, and the $0_{2 \times 2}$ blocks in the middle-left and middle-right. Red arrows point to the $0_{2 \times 2}$ blocks in the middle-left and middle-right.

$$B = \begin{bmatrix} \frac{1}{m_1} & 0 & 0 \\ 0 & \frac{1}{m_2} & 0 \\ 0 & 0 & \frac{1}{m_3} \end{bmatrix}, u = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix}, C = \begin{bmatrix} 1 & 0 & 0_{1 \times 2} & 0_{1 \times 2} \\ 0_{1 \times 2} & 1 & 0 & 0_{1 \times 2} \\ 0_{1 \times 2} & 0_{1 \times 2} & 1 & 0 \end{bmatrix}, y = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$$

Handwritten annotations: Red boxes highlight the B matrix and the u vector. Red arrows point to the u vector and the C matrix.

Examples: CARTS ON A RAILWAY CONNECTED THROUGH SPRINGS AND DAMPERS

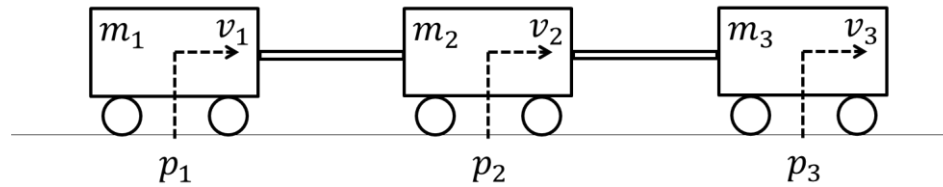


$$B_1 = \begin{bmatrix} 0 \\ 1 \\ \frac{1}{m_1} \\ 0_{2 \times 1} \\ 0_{2 \times 1} \end{bmatrix}, B_2 = \begin{bmatrix} 0_{2 \times 1} \\ 0 \\ 1 \\ \frac{1}{m_2} \\ 0_{2 \times 1} \end{bmatrix}, B_3 = \begin{bmatrix} 0_{2 \times 1} \\ 0_{2 \times 1} \\ 0 \\ 1 \\ \frac{1}{m_3} \end{bmatrix}$$

$$C_1 = [1 \quad 0 \quad 0_{1 \times 2} \quad 0_{1 \times 2}], C_2 = [0_{1 \times 2} \quad 1 \quad 0 \quad 0_{1 \times 2}],$$

$$C_3 = [0_{1 \times 2} \quad 0_{1 \times 2} \quad 1 \quad 0]$$

Examples: RIGIDLY CONNECTED CARTS ON A RAILWAY



$$(m_1 + m_2 + m_3)\ddot{p}_i = (m_1 + m_2 + m_3)\dot{v}_{\text{tot}} = -h_{\text{tot}}v_{\text{tot}} + u_1 + u_2 + u_3$$

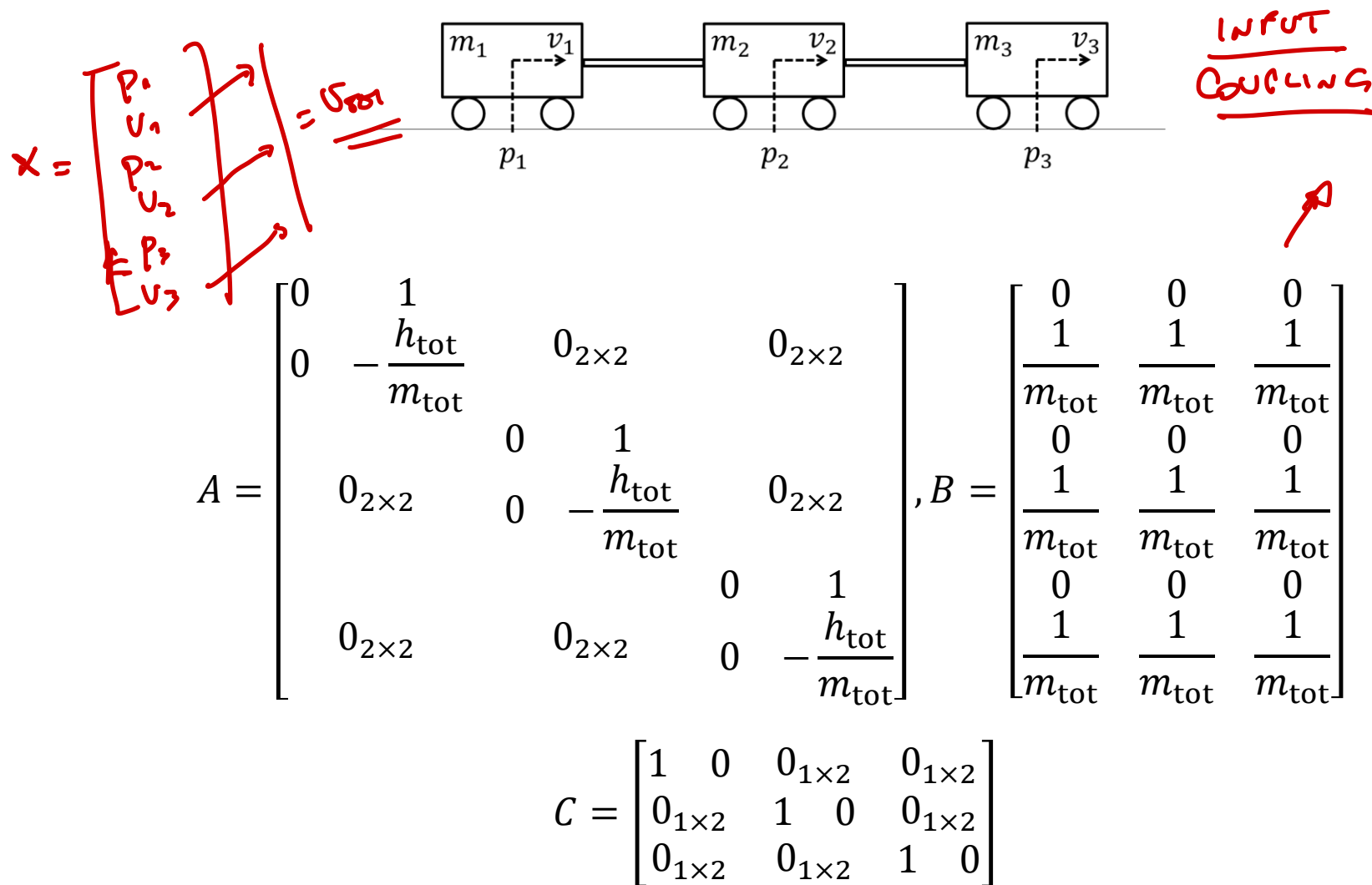
$$v_1 = v_2 = v_3 = v_{\text{tot}}$$

$$x = \begin{bmatrix} p_1 \\ p_2 \\ p_3 \\ v_{\text{tot}} \end{bmatrix}, A = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & -\frac{h_{\text{tot}}}{m_{\text{tot}}} \end{bmatrix}, B = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ \frac{1}{m_{\text{tot}}} & \frac{1}{m_{\text{tot}}} & \frac{1}{m_{\text{tot}}} \end{bmatrix}, C = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

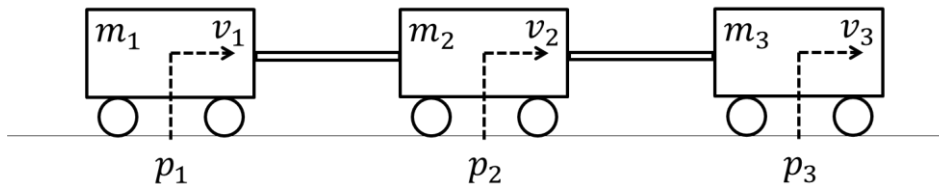
$$B_1 = \begin{bmatrix} 0 \\ 0 \\ 0 \\ \frac{1}{m_{\text{tot}}} \end{bmatrix}, B_2 = \begin{bmatrix} 0 \\ 0 \\ 0 \\ \frac{1}{m_{\text{tot}}} \end{bmatrix}, B_3 = \begin{bmatrix} 0 \\ 0 \\ 0 \\ \frac{1}{m_{\text{tot}}} \end{bmatrix}$$

$$C_1 = [1 \ 0 \ 0 \ 0], C_2 = [0 \ 1 \ 0 \ 0], \\ C_3 = [0 \ 0 \ 1 \ 0]$$

Examples: RIGIDLY CONNECTED CARTS ON A RAILWAY



Examples: RIGIDLY CONNECTED CARTS ON A RAILWAY

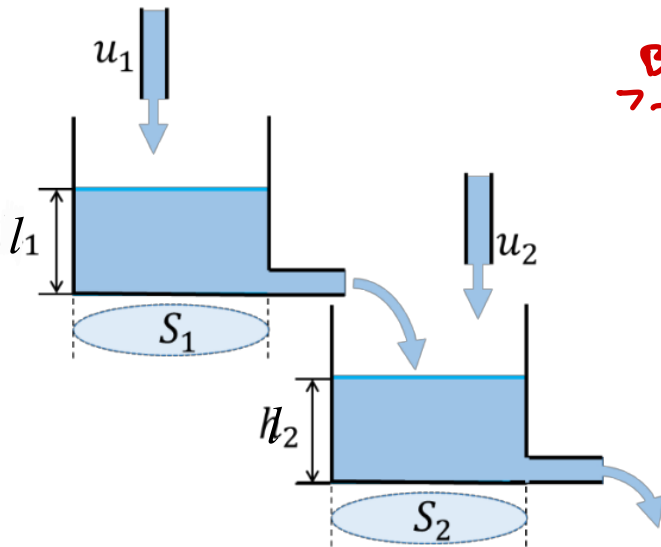


$$B_1 = \begin{bmatrix} 0 \\ 1 \\ \hline m_{\text{tot}} \\ 0 \\ 1 \\ \hline m_{\text{tot}} \\ 0 \\ 1 \\ \hline m_{\text{tot}} \end{bmatrix}, B_2 = \begin{bmatrix} 0 \\ 1 \\ \hline m_{\text{tot}} \\ 0 \\ 1 \\ \hline m_{\text{tot}} \\ 0 \\ 1 \\ \hline m_{\text{tot}} \end{bmatrix}, B_3 = \begin{bmatrix} 0 \\ 1 \\ \hline m_{\text{tot}} \\ 0 \\ 1 \\ \hline m_{\text{tot}} \\ 0 \\ 1 \\ \hline m_{\text{tot}} \end{bmatrix}$$

$$C_1 = [1 \quad 0 \quad 0_{1 \times 2} \quad 0_{1 \times 2}], C_2 = [0_{1 \times 2} \quad 1 \quad 0 \quad 0_{1 \times 2}],$$

$$C_3 = [0_{1 \times 2} \quad 0_{1 \times 2} \quad 1 \quad 0]$$

Examples: CASCADED LINEARIZED WATER TANKS



BLOCK-TRIANGULAR MATRIX

$$\begin{aligned} S_1 \dot{l}_1 &= -k_1 l_1 + u_1 \\ S_2 \dot{h}_2 &= -k_2 h_2 + k_1 l_1 + u_2 \end{aligned}$$

$$x = \begin{bmatrix} l_1 \\ l_2 \end{bmatrix}, A = \begin{bmatrix} -\frac{k_1}{S_1} & 0 \\ \frac{k_1}{S_2} & -\frac{k_2}{S_2} \end{bmatrix}, u = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}, B = \begin{bmatrix} \frac{1}{S_1} & 0 \\ 0 & \frac{1}{S_2} \end{bmatrix}, y = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}, C = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$B_1 = \begin{bmatrix} \frac{1}{S_1} \\ 0 \end{bmatrix}, B_2 = \begin{bmatrix} 0 \\ \frac{1}{S_2} \end{bmatrix}, C_1 = [1 \quad 0], C_2 = [0 \quad 1]$$